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|Research Article|

Enhancing Ultraviolet Communication Systems Using Machine Learning

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Abstract- Ultraviolet communication (UVC) has become a promising technology in secure and non-line of sight (NLOS) wireless communication particularly in complicated and blocked environments. But, UVC systems are very sensitive to atmospheric factors like fog, dust, and turbulence that greatly impair the quality of the signals, and the reliability of the system. This paper will suggest a machine learning-based solution to the modeling and prediction of the performance of UVC systems in different environments. A simulation model is created to produce artificial data, which depends on the most important input parameters, such as transmission distance, fog density, dust concentration, and signal power. The regression model used is the random forest regression, which predicts the most crucial performance parameters e.g. Bit Error Rate (BER) and Signal-to-Noise Ratio (SNR). The findings indicate that the prediction accuracy is high, and the Mean Squared Error (MSE) is low, indicating the suitability of the proposed method in modeling the nonlinear dynamics of the UVC channel. Machine learning integration offers greater system adaptability, less dependence on traditional analysis models, and offers a solid solution to enhance the performance and reliability of UVC systems in the real world.

Keywords: Ultraviolet Communication (UVC), Machine Learning, Non-Line-of-Sight (NLOS).

1. Introduction

In recent years, telecommunications networks have witnessed unprecedented growth in data traffic as a result of the expansion of electronic devices and services. This has led to the need to establish intelligent network systems that rely on data analysis and leveraging the behavior of different communication systems to meet the requirements of future devices and communication services(Ahmad et al., 2020).

Ultraviolet radiation is defined as electromagnetic radiation with wavelengths ranging from approximately 40 to 400 nanometers(Koller, 1965), Ultraviolet Communications (UVC) system have revolutionized modern communications as a smart alternative to traditional methods(Guo et al., 2021). Furthermore, ultraviolet communications have demonstrated high transmission rates and security capabilities, leading to their emergence as a key focus in modern communications research(Zhang, 2025).

Based on our vision that ultraviolet communication systems face many challenges and lack effective integration into the promising artificial intelligence revolution, some research questions were raised within the framework of employing machine learning in modern communication systems and enhancing ultraviolet communication systems.

2. Research Questions:

- How do environmental factors such as transmission distance, fog density, and dust concentration affect the performance of UVC systems?
- Can machine learning models accurately predict key performance metrics such as Signal-to-Noise Ratio (SNR) and Bit Error Rate (BER)?

- How effective is the proposed machine learning approach compared to traditional analytical models in modeling UVC channels?

3. Literature Review

3.1. Background

Ultraviolet communication (UVC) is a concept that can be traced to the beginning of the 20th century. It was introduced in the 1930s where Hulburt of the Naval Research Laboratory, summarized research done between 1926 and 1933, and published a report entitled: Signaling and Detection with Ultra-Violet and Infra-Red Radiation(Harvey, 1964).

Thereafter, there was a lot of theoretical and experimental development during the 1960s to 1990s. In 1964, G. L. Harvey suggested some of the technical problems that were involved in ultraviolet communication(Harvey, 1964). This was succeeded by D. E. Sunstein in 1968 who built the first experimental base of the UVC systems(Sunstein, 1968). Subsequently, D. M. Reilly proposed in 1976 the single scattering model to study the effects of pulse broadening in non-line-of-sight (NLOS) ultraviolet communication systems(Reilly, 1976). A relatively mature UVC system was first introduced in 1985, with the transmitter being a mercury lamp and the receiver being a photomultiplier tube (PMT) with a filter, by M. Geller et al., who were able to demonstrate the feasibility of ultraviolet communication(Geller et al., 1985). Luettgen (1991) enhanced the single scattering model by Reilly in 1991 to give it greater accuracy in its analysis(Luettgen,1991).

As the 21st century started, the number of studies in the area of UVC has grown considerably, especially as miniaturized and more efficient systems have become the focus. In 2000, GTE Corporation, in partnership with the U.S. military, came up with UVC system that can support a bit rate of 4.8 kbps with a rate of (106) bit error rate (BER) in a 2 km NLOS connection(Xu, 2000). Similarly, a research group at MIT Lincoln Laboratory, led by Professor Gary A. Shaw, was the first to use UV LEDs to construct experimental platforms, a shift of the traditional gas discharge lamps, with their large size, slow response time, and lack of wavelength selectability, to semiconductor-based sources of light(Yuan and Ma, 2016).

To facilitate these developments, in 2002 the Defense Advanced Research Projects Agency (DARPA) initiated the Semiconductor UV Optical Source (SUVOS) program and then the Deep Ultraviolet Avalanche Photodiode (DUVAP) program in 2007. These efforts were meant to come up with improved ultraviolet sources and detectors which were going to enhance more research in this area.

Tang et al. (2007) suggested a better single scattering model that could be effectively used to estimate the transmission distances in NLOS UVC systems in the solar-blind region based on channel modeling and performance analysis(Tang et al., 2007). This model was experimentally confirmed and it was extended to determine signal-to-noise ratio (SNR) and bit error rate (BER). Shaw et al. experimentally showed a 200 bps communication link in distance of up to 100 meters when using an array of 240 deep-UV LEDs at 275 nm(Shaw et al., 2004).

It was further developed in the following years. The performance of communication over a 100-meter distance with multiple receivers was studied in 2015 on the basis of the single scattering model(El-Shimy and Hranilovic, 2015). In most cases, this model offers a good correspondence to the experimental data concerning short-range communication (usually, less than 200 meters). But in the case of medium- and long-range ultraviolet communication, models of multiple scattering using the Monte Carlo technique are usually used. In spite of these innovations, the impact of atmospheric turbulence has not been fully considered in the current channel models.

3.2. Benefits and Challenges of Ultraviolet Communications system:

With the discovery of ultraviolet (UV) radiation and the commencement of research into its potential for use in communications, and based on its reliance on atmospheric scattering, it was found that non line of sight(NLOS) communication could be achieved using UV technology(Yuan and Ma, 2016). Researchers concluded that this radiation could address the problems of limited radio frequency spectrum and communication interference(Chen et al., 2022). UV communication technology has garnered increasing attention due to its ability to operate without a direct line of sight and its capacity to reduce background noise. Therefore, this type of communication is considered the ideal option for secure and confidential communications, particularly in military, emergency, and challenging environments(Guo et al., 2021), The scattering and absorption of ultraviolet signals in the atmosphere leads to much attenuation. The signal strength reduces very rapidly as the distance of transmission increases, which of course limits the distance of communication. This feature improves the security of UV communication systems, where eavesdropping becomes a

challenge as opposed to the traditional communication technologies(Yuan and Ma, 2016)

Despite the significant advantages offered by ultraviolet communications, they also present numerous challenges Scattering is one of the factors that influence the propagation of the ultraviolet (UV) signals through the atmosphere by attenuation that causes distortion of the signal and high Bit Error Rate (BER). In real world ultraviolet communication (UVC) systems, various issues come up especially the degradation and attenuation of signals by atmospheric turbulence and scattering effects, which adversely affect the reliability of communication and the overall system performance.The existing literature has concentrated on simplified conditions of scenarios or narrow channel conditions only and there is a gap in literature that has dealt with more complex conditions, with extreme turbulence and multiple path interference(Maclure et al., 2022). Moreover, it is also a significant challenge to deliver real-time processing efficiency and at the same time ensure low computational complexity to make systems implementable in practice.Moreover, UVC signal transmission is very sensitive to the environmental factors, weather conditions, atmospheric changes, and brightness. This sensitivity restricts the strength and flexibility of existing models. Thus, more detailed methods are required, which would utilize different datasets that were obtained in different environmental conditions to improve the performance of the system and its generalization in real-life situations.(Zhang, 2025)

4. Ultraviolet comunication(UVC) and Machine Learning Integration-Proposed System:

Atmospheric scattering, turbulence, and attenuation are the environmental conditions that have a great impact on ultraviolet communication (UVC) systems. These aspects present nonlinear and complex effects making the communication channel a difficult task to model accurately.Conventional models of analysis are usually founded on simplified assumptions and predetermined equations, which restricts their capability to reflect the dynamic behavior of the real world. These models can therefore not make precise predictions of the performance particularly in different and erratic weather conditions. Intelligence in communication networks is defined as a system that strives to achieve its goals autonomously without human intervention under conditions of uncertainty(Stuart and Peter, 2010). Traditional ultraviolet (UV) communications require continuous human intervention to assess attenuation and dispersion, adjust settings based on experience, continuously monitor SNR, BER, and signal quality, intervene immediately in case of degradation, reset the system, correct errors, and improve or modify algorithms. Therefore, a non-autonomous system does not learn from its environment and relies, in part, on human intervention, especially in response to changes in environmental conditions.The idea of integrating machine learning into ultraviolet communication systems is based on the system interacting and adapting to environmental changes in the presence of a dynamic, changing environment affected by weather conditions and various circumstances, which in turn affect the quality of communication. The proposed system relies on learning from experience based on the information it collected during its operation. The need to create a system that integrates machine learning and ultraviolet communication arose for the following reasons: mitigating the risks related to human intervention, enabling networks to self-regulate and repair themselves with changes in data traffic and network, in addition to their dynamic requirements.

Machine learning has emerged as a powerful tool in communication systems due to its ability to model complex nonlinear relationships without requiring explicit mathematical formulations. By learning directly from data, machine learning models can capture hidden patterns and dependencies between input parameters and system performance (Ahmad et al., 2020).

Based on these capabilities, integrating machine learning into UVC systems provides a promising approach to overcome the limitations of traditional methods. In this work, environmental parameters such as transmission distance, fog density, dust concentration, and signal power are utilized as input features, while system performance metrics such as Bit Error Rate (BER) and Signal-to-Noise Ratio (SNR) are considered as output variables.The machine learning model learns the mapping between these inputs and outputs, enabling accurate prediction of system performance under different environmental conditions. This data-driven approach enhances the adaptability and flexibility of UVC systems and provides a more robust alternative to conventional analytical modeling techniques.

Judging by these capabilities, machine learning can be introduced into UVC systems, which can be seen as a promising method of overcoming the limitations of the traditional approaches. The input features in this work are the environmental parameters like the distance of transmission, the density of the fog, the concentration of dust and the signal power, whereas the output variables are the system performance parameters like the Bit Error Rate (BER) and the Signal-to-Noise Ratio (SNR).

The machine learning model is trained on the relationship between these inputs and outputs and can predict the system performance in various environmental conditions accurately. This data-driven approach enhances the adaptability and flexibility of UVC systems and provides a more robust alternative to conventional analytical modeling techniques.

5. Methodology:

This paper introduces an experimental architecture of machine learning to model and predict the performance of ultraviolet communication (UVC) systems under different atmospheric conditions. The methodology consists of four main stages: data generation, feature selection, model training, and performance evaluation.

5.1. Data Generation

An artificial simulation environment was created in order to create a dataset that simulates the behavior of a UVC system in various environmental conditions. To create realistic variations in operating conditions, a total of 500 samples (n = 500) were generated using uniform random distribution. The input features are transmission distance, fog density, dust concentration and transmission power. In particular, the distance values were set to [10, 100] to reflect the different separations between the transmitter and receiver, and the level of fog and dust set to [0, 1] to reflect the various levels of atmospheric attenuation. The transmission power was adjusted in [1, 10] in order to have different signal strength levels. These properties were chosen because they have major influence on the signal propagation in the ultraviolet channels.

5.2. System Modeling

The outputs of the system are Signal-to-Noise Ratio (SNR) and Bit Error Rate (BER), which are the most common metrics in assessing the quality and reliability of communication. The simplified physics-inspired formulation which models the SNR and reflects the inverse relationship of the quality of signal, distance, and environmental interference was as follows:

$$\text{SNR} = \frac{P * 120}{d * (1 + f + a)}$$

(1)

where P is transmission power, d is distance, and f and a represent fog and dust effects respectively. The constant 120 is used as a scaling factor to normalize output values for simulation purposes and does not represent a physical constant. This formulation reflects the general behavior of communication systems, where increasing power improves SNR, while increasing distance and atmospheric attenuation reduces it.

The Bit Error Rate (BER) was then derived using an exponential decay relationship:

$$\text{BER} = e^{-SNR}$$

(2)

This captures the general trend in communication systems where BER decreases as SNR increases

5.3. Machine Learning Model

Upon generation of the datasets, the data was divided into training and testing sets, to facilitate the appropriate testing of the model performance. The nonlinear correlations between input characteristics (power, distance, fog, dust) and output characteristics (SNR and BER) were learned with the help of a Random Forest regression model. This algorithm was selected due to its robustness and ability to capture complex nonlinear dependencies.

5.4. Performance Evaluation

Lastly, the performance of the models was measured with the test dataset comparing the predicted values to actual simulated values. Mean Squared Error (MSE) was used to evaluate the accuracy of prediction and the ability of the model to be generalized to actual outputs as well as comparing the predicted and actual outputs visually.

6. Transition to Simulation

6.1. BER prediction

A simulation-based approach was used to prove the efficiency of the proposed integration. The simulation offers a controlled condition that can simulate the UVC system behavior and produce data to train and evaluate the UVC system. To model the correlation between the environmental conditions and the system performance, a synthetic dataset was created with references to the developed simulation model.

Some of the input features included distance, fog, dust and signal power and output variables were BER and SNR. Using Python, a Random Forest regression model was used to train on the generated data.

The data was split into test and training data, and the model was assessed on the capacity to forecast the value of BER in various environmental conditions.

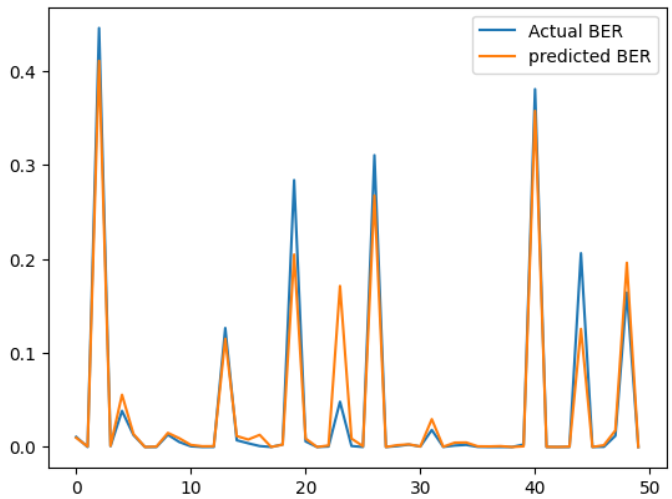

Fig. 1. BER Prediction

The implemented model achieved a Mean Squared Error (MSE) of 0.0004680276551369002, indicating high prediction accuracy.Comparison of the actual and predicted values of BER indicates that there is a high correlation between the two results, which depicts the efficiency of the machine learning model in representing the relationship between environmental conditions and signal degradation.

6.2. SNR prediction

The data was also analyzed to determine the ability of the suggested model to predict Signal-to-Noise Ratio (SNR) in different environmental conditions. The findings show that the predicted SNR values are closely correlated to the overall trend of the actual measured values, which is evidence that the model has a great learning capacity and can capture an underlying relationship between the input features and the resultant SNR.

Though there are some insignificant deviations between the predicted and actual values, the deviations are not considerable in various environmental situations. This stability shows the strength of the model under different conditions like the distance variation, atmospheric conditions, and signal strength.

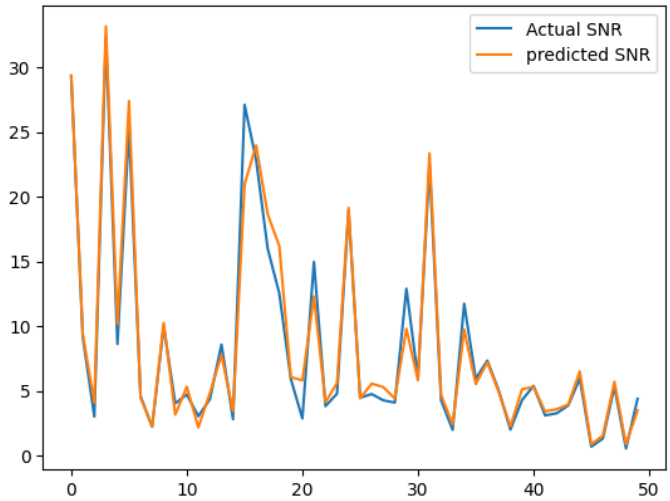

Fig. 2. SNR Prediction

In addition, the Mean Squared Error (MSE) was estimated to be about 5.50 which indicates a reasonable degree of prediction accuracy given the inherent variability and uncertainty inherent in the simulated environment. The comparatively small error value also helps to confirm the accuracy of the model predictions.

7. Discussion:

The findings confirm the machine learning model has been able to learn the nonlinear nature of the UVC channel. The great precision obtained indicates that data-driven modeling is effective in forecasting system performance.It was possible to test the proposed approach in controlled conditions with the help of simulation. Nevertheless, the use of synthetic data can restrict the capability to depict real-life settings in full.

8. Recommendations:

This research underscores the need for studies into additional technology techniqes that can integratre with ultraviolet comunication to enhancing the quality of comunication. and develop a comprehensive and flexible conceptual framework that not only promotes technology integration but also adapts to rapid and ever-evolving technological progress. The pace of innovation and development requires a comprehensive and constant examination of all relevant factors that have effect on the ultraviolet signals. This approach will ensure that our methodologies and strategies remain cutting-edge and highly effective in promoting technology integration with modern comunications systems. By doing so, we can better equip comunications to succeed in an increasingly digital and interconnected world, and address the multifaceted challenges and opportunities that arise from this rapid evolution and of the unpreditable enviroment.

9. Conclusion

This paper presented a machine learning-based structure that can help improve the prediction of the performance of ultraviolet communication systems under different atmospheric conditions. The analysis revealed that the conventional forms of analysis are weak in the depiction of the nonlinear and complicated behavior of the UVC channels particularly in dynamic contexts. Through a data-driven method based on the Random Forest regression, the proposed model was able to learn the connection between environmental parameters and performance metrics of the system (BER and SNR).

The results of the simulation demonstrated that the machine learning models had high predictive accuracy and are able to predict signal degradation and communication quality. This will enhance system flexibility and minimize reliance on manual tuning and simplified assumptions. Although the results are promising, synthetic data is still used, and the future research must concentrate on model validation on real-world data and consider the use of more sophisticated machine learning methods.

Altogether, the idea to incorporate machine learning into UVC systems is a major milestone towards creating intelligent, and reliable communication systems, which can learn and work effectively in the harsh environmental conditions.

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